

2023

Answer No. 21

1. Statistics plays a crucial role in computer applications, enabling data-driven decision-making analysis, and innovation. Here's a brief discussion of its applications:

- a. Data Analysis and Visualization
- b. Machine Learning and Artificial Intelligence
- c. Data Mining
- d. Natural Language processing
- e. Software engineering and testing.

Answer No. 12

Statistics is a branch of mathematics that deals with the collection, organization, analysis, interpretation, and presentation of data.

x	f	fx	cf
0	15	0	15
1	22	22	37
2	28	56	65
3	35	105	100
4	42	168	142
5	34	170	176
6	24	144	200
$N = 200$		$\Sigma fx = 665$	

$$\text{Mean } (\bar{x}) = \frac{\Sigma fx}{N} = \frac{665}{200} = 3.325$$

$$\begin{aligned} \text{Position of median} &= \left(\frac{N+1}{2} \right)^{\text{th}} \text{ item} \\ &= \left(\frac{200+1}{2} \right)^{\text{th}} \text{ item} \\ &= 100.5^{\text{th}} \text{ item} \end{aligned}$$

Here, cf just equals 200. It's corresponding value is 3.
Hence median = 3

Here, maximum frequency is 42 which is corresponding to

4.
mode = 4.

Answer No. 48

Correlation refers to the statistical relationship or association between two variables.

X	Y	X ²	Y ²	XY
2	10	4	100	20
4	9	16	81	36
6	8	36	64	48
8	7	64	49	56
10	6	100	36	60
$\Sigma X = 30$	$\Sigma Y = 40$	$\Sigma X^2 = 220$	$\Sigma Y^2 = 330$	$\Sigma XY = 220$

$n = 5$

we use the formula;

$$r = \frac{n \Sigma XY - (\Sigma X)(\Sigma Y)}{\sqrt{[n(\Sigma X^2 - (\Sigma X)^2/n)] [n(\Sigma Y^2 - (\Sigma Y)^2/n)]}}$$

Putting values;

$$r = \frac{5(220) - (30)(40)}{\sqrt{[5(220) - (30)^2][5(330) - (40)^2]}}$$

$$r = \frac{1100 - 1200}{\sqrt{[100 - 900][1650 - 1600]}}$$

$$r = \frac{-100}{\sqrt{(200)(50)}}$$

$$r_{xy}, r = \frac{-100}{\sqrt{10000}}$$

$$r_{xy}, r = \frac{-100}{100}$$

$$r_{xy}, r = -1$$

∴ Correlation Coefficient (r) = -1

- Interpretation of correlation coefficient
- The correlation coefficient $r = -1$ indicates a perfect negative linear relationship between the number of study hours and the number of sleeping hours. As study hours increase, sleeping hours decrease in a perfectly consistent manner.

Compute Coefficient of Determination

$$r^2 = (-1)^2 = 1$$

Interpretation:

- The coefficient of determination (r^2) is 1, or 100%.
- This means that 100% of the variation in sleeping hours can be explained by the variation in study hours.

Answer No. 14

Regression is a statistical method used to model the relationship between a dependent variable and one or

new Independent variable.

X	Y	X^2	Y^2	XY
56	189	3364	35721	10962
69	225	4761	55225	16215
43	193	1849	37249	8299
39	177	1521	31329	6903
63	154	3969	23716	9702
52	191	2704	36481	9932
47	213	2209	45369	10011
31	169	961	27225	5115
74	198	5476	39204	14652
36	182	1296	32761	6516
$\Sigma X = 512$	$\Sigma Y = 1896$	$\Sigma X^2 = 28110$	$\Sigma Y^2 = 364280$	$\Sigma XY = 9837$

1) Let the regression equation of y on x be
 $y = a + bx \quad (i)$

where,

y = dependent variable

X = independent variable

a = y -intercept

b = regression coefficient

Here, a and b are constants and need to determine by solving following linear equations obtained from method of least square.

$$\Sigma y = na + b \Sigma x$$

$$\text{or, } 1896 = 10a + b 512$$

$$\text{or, } 10a + 512b = 1896 \quad (ii)$$

$$\sum xy = ax + b \sum x^2$$

$$\text{or, } 983.7 = a \cdot 512 + b \cdot 28110$$

$$\text{or, } 512a + 28110b = 983.7 \quad \dots (i)$$

Multiplying eqn (i) by 512 and eqn (ii) by 10 and subtract

eqn (i) from (ii)

$$51209a + 262144b = 970752$$

$$- 5120a + 281100b = 983070$$

$$\hline 189566b = 712318$$

$$\text{or, } b = \frac{712318}{189566} = 0.649$$

$$189566$$

from eqn (i)

$$102 + 512b = 1896$$

$$\text{or, } 102 + 332.288 = 1896$$

$$\text{or, } 102 = 1563.712$$

$$\text{or, } a = 156.37$$

Hence required regression eqn is

$$\hat{y} = 156.37 + 0.649x$$

$$\text{ii. } \hat{y} = 156.37 + 0.649(60) = 156.37 + 38.94 = 195.31$$

iii. Interpretation of regression coefficient;

The slope 0.649 indicates that for each additional year of age, cholesterol level increases by

$$\binom{15}{12} = \frac{15!}{12!(3!)}$$

about 0.699 mg/dl.

Answer No. 15

A discrete random variable X is said to follow Binomial distribution if it assumes only non-negative values and its probability mass function (pmf) is,

$$P(X=x) = \binom{n}{x} p^x q^{n-x}, \quad x=0, 1, 2, \dots, n$$

0 otherwise

$$n=15$$

$p=0.7$ is probability of success

$q=1-p=0.3$ is probability of failure.

We use the binomial probability formula:

$$P(X=x) = \binom{n}{x} p^x q^{n-x}$$

$$?) \quad P(X \geq 12) = P(X=12) + P(X=13) + P(X=14) + P(X=15)$$

$$= \binom{15}{12} (0.7)^{12} (0.3)^3 + \binom{15}{13} (0.7)^{13} (0.3)^2$$

$$+ \binom{15}{14} (0.7)^{14} (0.3)^1 + \binom{15}{15} (0.7)^0 (0.3)^0$$

$$360360 \quad \frac{15}{1312}$$

$$= 455 \times 0.01384 \times 0.027 + 0.091 + 0.0305 + 0.00475$$

$$= 0.296$$

$$\begin{aligned} \text{ii) } P(X=20) &= {}^{15}C_2 (0.7)^{10} (0.3)^5 \\ &= 30.3 \times 0.02825 \times 0.00243 \\ &\approx 0.206 \end{aligned}$$

There's about 20.6 % chance that exactly 10 out of 15 chips will have thick enough coatings.

Answer No. 26

Given:

- $SD(\sigma) = 0.95$ seconds
 - Confidence level = 95%
 - Margin of Error (E) = 0.01 second
 - $Z = 1.96$ score for 95% confidence level
- Sample size formula,

$$n = \left(\frac{Z \cdot \sigma}{E} \right)^2$$

$$\text{or, } n = \left(\frac{1.96 \cdot 0.95}{0.01} \right)^2$$

$$\text{or, } n = \left(\frac{1.862}{0.01} \right)^2$$

$$\approx 34670.44$$

Since sample size must be a whole number, we round up:

$$n = 34671$$

To be 95% confident that the error in estimating the mean reaction time will not exceed 0.01 seconds, the psychologist must take a sample of at least 34,672 measurements.

Answer No. 17

Sampling is the process of selecting a subset from a larger group (called the population) in order to make inferences or draw conclusions about the population as a whole.

Given population:

$\{1, 3, 4, 8\}$

i) The total number of sub samples,
$$\binom{4}{2} = 6$$

to the possible samples of size 2 are:

Sample no.	Sample
1	(1, 3)
2	(1, 4)
3	(1, 8)
4	(3, 4)
5	(3, 8)
6	(4, 8)

ii) Step 1: Compute the population mean

$$\mu = \frac{1+3+4+8}{4} = 4$$

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Step 2: Compute mean of each sample

Sample

(1, 3)

(1, 4)

(1, 8)

(3, 4)

(3, 8)

(4, 8)

sample mean

$$(1+3)/2 = 2$$

$$(1+4)/2 = 2.5$$

4, 5

3, 5

5, 5

6

Step 3: Compute the expected value
(mean of sample means)

$$\text{Average of sample mean} = 2 + 2.5 + 4.5 + 3.5 + 5.5 + 6$$

$$= 4$$

This equals the population mean $\mu = 4$

Since, the average of all possible sample means equals the population mean, we conclude that:

The sample mean is an unbiased estimator of the population mean.

Answer No. 18

∴ Wage bill = (Number of workers) ×
(Average monthly wage)

From A:

$$500 \times 186 = 93000$$

For B:
600 × 275 = 165,000

For B has a larger bill (Rs 1,05,000)
Rs 93,000

Q. To compare variability, we use the CV:

$$CV = \frac{SD}{\text{mean}} \times 100$$

$$\sigma = \sqrt{\text{variance}}$$

For Firm A:

$$\text{Variance} = 81 \quad SD = \sqrt{81} = 9$$

$$\text{mean} = 186$$

$$CV = \frac{9}{186} \times 100 \approx 4.84\%$$

For Firm B:

$$\text{Variance} = 203 \quad SD = \sqrt{203} \approx 14.24$$

$$\text{Mean} = 275$$

$$CV = \frac{14.24}{275} \times 100 \approx 5.17\%$$

Answer:

Firm B has greater variability in individual wage, as its CV is higher.

iii) Let's denote:

$$n_1 = 500, \bar{x}_1 = 186, s_1^2 = 81$$

$$n_2 = 600, \bar{x}_2 = 175, s_2^2 = 100$$

$$\text{Total number of words: } N = n_1 + n_2 = 1100$$

• Combined Mean

$$\bar{x}_{\text{combined}} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

$$= \frac{500 \cdot 186 + 600 \cdot 175}{1100}$$

$$= \frac{500 \cdot 186 + 600 \cdot 175}{1100}$$

$$= 180$$

• Combined Variance:

Use the formula:

$$s^2_{\text{combined}} = \frac{n_1 [(s_1^2 + d_1^2)] + n_2 [(s_2^2 + d_2^2)]}{n_1 + n_2}$$

where,

$$d_1 = \bar{x}_1 - \bar{x}_{\text{combined}} = 186 - 180 = 6$$

$$d_2 = \bar{x}_2 - \bar{x}_{\text{combined}} = 175 - 180 = -5$$

$$s_1^2 + d_1^2 = 81 + 36 = 117$$

$$s_2^2 + d_2^2 = 100 + 25 = 125$$

$$s^2_{\text{combined}} = \frac{500 \cdot 117 + 600 \cdot 125}{1100}$$

$$= 121.36$$

Answer No. 19

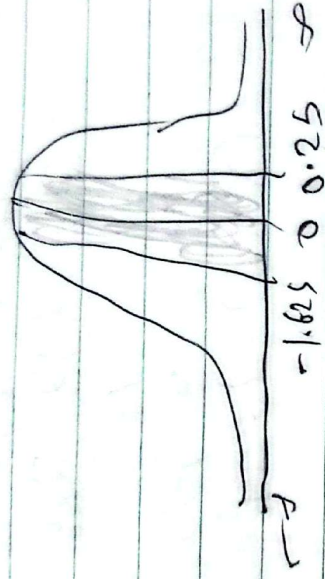
$$\text{mean}(\mu) = 266$$

$$\text{sd}(\sigma) = 26$$

If X follows normal distribution then
standard normal variate of X is,

$$Z = \frac{X - \mu}{\sigma}$$

$$Z = \frac{X - 266}{26}$$



$$i) P(240 < X < 270)$$

$$\text{When } X = 240$$

$$Z = \frac{240 - 266}{26}$$

$$= -1.625$$

$$\text{When } X = 270$$

$$Z = \frac{270 - 266}{26}$$

$$= 0.25$$

$$P(240 < X < 270) = P(-1.625 < Z < 0.25)$$

$$P(-1.625 < Z < 0.25)$$

$$= P(-1.625 < Z < 0) + P(0 < Z < 0.25)$$

$$= 0.4474 + 0.0987$$

$$= 0.5461$$

$$ii) P(X > 275)$$

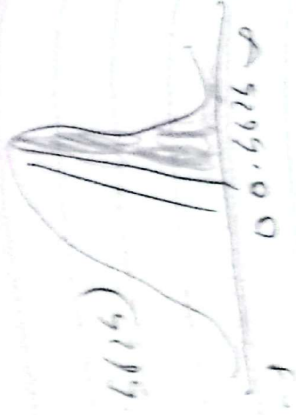
$$\text{When } X = 275$$

$$Z = \frac{275 - 266}{26}$$

$$= 0.3462$$

$$= 0.5625$$

$$P(X > 275) = P(Z > 0.3462)$$

$$= P(Z < 0) + P(0 < Z < 0.6626) \\ = 0.5 + 0.243 \\ = 0.743$$


$$(ii) P(X < 272)$$

$$\text{where } X = 272$$

$$Z = \frac{272 - 266}{10}$$

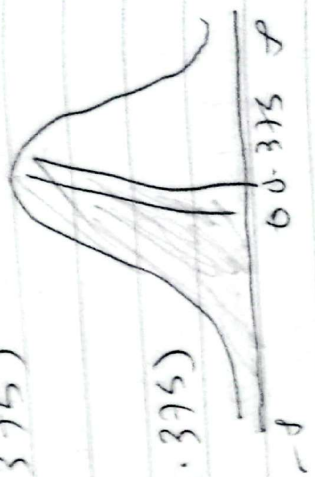
$$= 0.375$$

$$P(X < 272) = P(Z < 0.375)$$

$$P(Z < 0.375)$$

$$= P(-Z < 0) + P(0 < Z < 0.375)$$

$$= 0.5 + 0.1443$$

$$= 0.6443$$


Answer No. 2.

ANOVA (Analysis of variance) is a statistical method used to compare the means of three or more groups to determine whether there are statistically significant differences among them.

Types of ANOVA:

- One-way ANOVA: Used when data is categorized.
- Two-way ANOVA: Involves two factors and can also test for interaction between them.

Step 1: Setting up Hypothesis:
 Null Hypothesis (H_0) = $\mu_A = \mu_B = \mu_C$
 There is no significant difference in the sample means.

Alternative Hypothesis (H_1) = $\mu_A \neq \mu_B \neq \mu_C$
 There is significant difference in the sample means.

Step 2: Test statistics:
 Under H_0 for one way ANOVA, the test statistic is:

$$F_{cal} = \frac{MSC}{MSE}$$

$$MSC = \frac{SSC}{k-1}$$

$$MSE = \frac{SSE}{n-k}$$

S.N	Sample I (x_i)	x_2	x_3	$(x_i)^2 \cdot (x_2)^2 / (x_3)^2$	
				$(x_1)^2$	$(x_2)^2$
1	4	7	4	16	49
2	5	5	8	25	25
3	6	6	9	36	36
4	5	5	2	25	8
				25	4
				25	4

$T_1 = \sum x_1 = 20$ $T_2 = \sum x_2 = 28$ $T_3 = 23$ $\sum x_1^2 = 135$ $\sum x_2^2 = 165$

Grand Total (T) = $T_1 + T_2 + T_3$

$$= 20 + 28 + 23 = 71$$

$$\begin{aligned}\text{Correction factor (C.F.)} &= \frac{T^2}{n} \\ &= \frac{(66)^2}{12} \\ &= 363\end{aligned}$$

$$\begin{aligned}\text{Total Sum of Square (TSS)} &= \sum x_1^2 + \sum x_2^2 + \sum x_3^2 \\ &\quad - (C.F.) \\ &= 102 + 135 + 165 - 363 \\ &= 39\end{aligned}$$

Sum of square due to column (SSC) =

$$\begin{aligned}&= \sum \left(\frac{T_j^2}{n_j} \right) - (C.F.) \\ &= \frac{T_1^2}{n_1} + \frac{T_2^2}{n_2} + \frac{T_3^2}{n_3} - (C.F.) \\ &= \frac{20^2}{4} + \frac{23^2}{4} + \frac{23^2}{4} - 363 \\ &= 100 + 132.25 + 132.25 - 363 \\ &= 1.5\end{aligned}$$

$$\begin{aligned}\text{Sum of squares due to factors (SSE)} &= TSS - SSC \\ &= 39 - 1.5 \\ &= 37.5\end{aligned}$$

one-way ANOVA Table

Source of variation	Sum of Squares (S.S)	Degree of freedom (D.F)	Mean Sum of Squares (MS)	F-ratio
Between the samples (column)	$SSC = 1.5$	$k - 1 = 3 - 1 = 2$	$MSC = \frac{SSC}{k - 1} = \frac{1.5}{2} = 0.75$	$F_{cal} = \frac{MSC}{MSE} = \frac{0.75}{4.16} = 0.18$
Within the samples (rows)	$SSE = 37.5$	$n - k = 12 - 3 = 9$	$MSE = \frac{SSE}{n - k} = \frac{37.5}{9} = 4.16$	
Total	$TSS = 39$	$n - 1 = 12 - 1 = 11$		

Step 3: Level of significance, α :

Here, the level of significance is 0.05
or 5%.

$$\alpha = 0.05$$

Step 4: Critical Value
Tabulated value of F at 5% level of significance for (2, 9) is:

$$F_{tab} = 4.2565$$

Step 5: Decision

Since $F_{cal} < F_{tab}$, H_0 is accepted and H_1 is rejected.

Step 6: Conclusion
There is no statistically significant difference between the

means of three samples at 5% level of significance.

i.e. The means of three samples are not significantly different.